SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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Your Code

```java
import java.util.*;

public class SmartHospitalPatientRegistrationSystem {

    static class Patient {
        String patientId;
        String patientName;
        String mobileNumber;
        String department;
        String doctorName;

        Patient(String patientId, String patientName, String mobileNum
            this.patientId = patientId;
            this.patientName = patientName;
            this.mobileNumber = mobileNumber;
            this.department = department;
            this.doctorName = doctorName;
        }
    }

    static ArrayList<Patient> patientList = new ArrayList<>();
    static HashMap<String, Patient> patientMap = new HashMap<>();
    static HashMap<String, Integer> departmentCount = new HashMap<>();

    public static void main(String[] args) {
        String[][] inputPatients = {
                {"P101", "raHUl kumAr", "9876543210", "Cardiology", "D
                {"P102", "   Priya   ", "9876543211", "Ortho", "Dr.Mee
        };

        for (String[] p : inputPatients) {
            registerPatient(p[0], p[1], p[2], p[3], p[4]);
        }

```

Input

```
stdin input
```

Output

```
Patient Registered Successfully.
Patient Registered Successfully.
Patient Details
ID : P101
Name : Rahul Kumar
Mobile : 9876543210
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

0.00 / 20 marks

```java
        String searchId = "P101";
        searchPatient(searchId);

        displayAllPatients();

        countDepartmentWisePatients();
    }

    static void registerPatient(String patientId, String patientName,

        if (patientList.size() >= 100) {
            System.out.println("Cannot register more than 100 patients
            return;
        }

        patientId = trimAll(patientId);
        patientName = trimAll(patientName);
        mobileNumber = trimAll(mobileNumber);
        department = trimAll(department);
        doctorName = trimAll(doctorName);

        if (patientMap.containsKey(patientId)) {
            System.out.println("Duplicate Patient ID. Registration fai
            return;
        }

        if (!isAlphabeticWithSpaces(patientName)) {
            System.out.println("Invalid Patient Name. Registration fai
            return;
        }

        if (!mobileNumber.matches("\\d{10}")) {
            System.out.println("Invalid Mobile Number. Registration fa
            return;
        }
```

```
        String deptUpper = department.toUpperCase();
        String properName = toProperCase(patientName);

        Patient patient = new Patient(patientId, properName, mobileNum
        patientList.add(patient);
        patientMap.put(patientId, patient);

        System.out.println("Patient Registered Successfully.");
    }

    static void searchPatient(String patientId) {
        patientId = trimAll(patientId);
        Patient p = patientMap.get(patientId);

        if (p == null) {
            System.out.println("Patient Not Found");
        } else {
            System.out.println("Patient Details");
            System.out.println("ID : " + p.patientId);
            System.out.println("Name : " + p.patientName);
            System.out.println("Mobile : " + p.mobileNumber);
            System.out.println("Department : " + p.department);
            System.out.println("Doctor : " + p.doctorName);
            System.out.println("Name Length : " + p.patientName.replac
        }
    }

    static void displayAllPatients() {
        System.out.println("-----------------------------------------
        System.out.println("ID Name Mobile Department Doctor");
        System.out.println("-----------------------------------------
        for (Patient p : patientList) {
            System.out.println(p.patientId + " " + p.patientName + " "
        }
        if (patientList.isEmpty()) {
```

```java
        }
        if (patientList.isEmpty()) {
            System.out.println("No patients available.");
        }
    }

    static void countDepartmentWisePatients() {
        departmentCount.clear();

        for (Patient p : patientList) {
            departmentCount.put(p.department, departmentCount.getOrDef
        }

        System.out.println("Department Count");
        for (Map.Entry<String, Integer> e : departmentCount.entrySet()
            System.out.println();
            System.out.println(e.getKey() + " : " + e.getValue());
        }
    }

    static String trimAll(String s) {
        return s == null ? "" : s.trim().replaceAll("\\s+", " ");
    }

    static boolean isAlphabeticWithSpaces(String name) {
        return name.matches("[A-Za-z]+( [A-Za-z]+)*");
    }

    static String toProperCase(String s) {
        s = trimAll(s);
        String[] parts = s.split(" ");
        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i].toLowerCase();
            if (w.length() > 0) {
                sb.append(Character.toUpperCase(w.charAt(0))).append(w
```

```
            System.out.println(e.getKey() + " : " + e.getValue());
        }
    }

    static String trimAll(String s) {
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        String[] parts = s.split(" ");
        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i].toLowerCase();
            if (w.length() > 0) {
                sb.append(Character.toUpperCase(w.charAt(0))).append(w
            }
            if (i != parts.length - 1) sb.append(" ");
        }
        return sb.toString();
    }
}
```

Run Save Submit